AthletixAI ��� README

AthletixAI

AI-powered sports talent assessment — using nothing but a phone camera.

🔗 **Live:** athletixai.onrender.com 👥 **Built by:** Archita Tripathi · Manthan Dhanraj

Record a training clip on any phone. On-device AI extracts your body pose, measures how you move, and returns a performance report scored across five metrics — tuned to your sport. No sensors, no lab, no cost.

The problem

India does not lack sporting talent. It lacks **measurement**.

- **No coach** — a rural athlete has no trained eye to tell them what to fix.
- **No baseline** — without numbers, nobody knows if they are improving or plateauing.
- **No visibility** — selectors can only pick from athletes they have physically seen.

A professional biomechanics lab costs lakhs and exists in a handful of cities, so most athletes are never measured at all. AthletixAI exists so that fewer athletes lose their chance for lack of resources.

What it does

An athlete records a drill on any phone (or uses live camera mode) and gets a report in seconds:

Metric	What it measures
Speed	Peak movement burst and step cadence
Agility	Body symmetry and direction-change frequency
Strength	Joint flexion depth and range of motion
Stamina	Sustained output across the session
Technique	Movement control, form quality, tracking confidence

Each report also gives sport-specific insights, corrective drills for the weakest metric, injury-watch zones, and a progress trend versus previous sessions.

How the AI actually works

This is a real computer-vision pipeline, not a chatbot wrapper.

1. Pose estimation (real ML). Google's MediaPipe Pose Landmarker runs in the browser and returns **33 body keypoints** per frame — each with an x, y and a visibility confidence.

2. Feature extraction (real measurement). From those keypoints the engine computes joint angles using the dot-product formula `cos(θ) = (v₁ · v₂) / (|v₁| × |v₂|)` — e.g. the knee angle from the hip → knee → ankle triangle — plus frame-to-frame velocity, left/right symmetry, step cadence and range of motion. Frames where a landmark's visibility drops below 0.3 are discarded, so a partly hidden body never pollutes the result.

3. Category-aware scoring. Different sports reward different things, so one formula would be wrong — an archer holding still is performing well, a sprinter standing still is not. The engine carries **six scoring profiles** (Precision, Endurance, Combat, Strength, Artistic, Dynamic). An *effort gate* scales the dynamic metrics, so low-effort movement cannot reach elite numbers.

It refuses to guess

A tool that invents a number when it cannot see the athlete is worse than no tool at all. The engine returns an honest error instead of a score when: fewer than 12 frames contain a detected person; the person is visible in under 40% of frames; there is not enough movement data; or the footage is too blurry (score is capped, not inflated). **There is no random number anywhere in the scoring path** — the same clip always produces the same score.

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35 sports, across 8 categories

Category	Count	Sports
Team	9	Football, Basketball, Volleyball, Rugby, Hockey, Handball, Lacrosse, Sepak Takraw, Kabaddi
Combat	7	Boxing, Wrestling, MMA, Judo, Karate, Taekwondo, Fencing
Bat & Ball	3	Cricket, Baseball, Softball
Racquet	3	Tennis, Badminton, Table Tennis
Endurance	2	Athletics, Cycling
Precision	5	Archery, Shooting, Golf, Snooker/Pool, Billiards
Artistic	5	Gymnastics, Parkour, Climbing, Skateboarding, Breakdancing
Strength	1	Weightlifting

13 sports were **deliberately removed** (swimming, diving, equestrian, chess and others) because a phone camera cannot honestly analyse them. Shipping unreliable scores for those would undermine the 35 that work.

Features by role

Athlete — live pose analysis with skeleton overlay · video upload & analysis · per-sport reports · progress tracking with AI verdicts · career roadmap & weekly plan · shareable athlete card & badges · coach directory, ratings, messaging · leaderboard · PDF export.

Coach — squad dashboard · full athlete report history · Recruiter Hub with AI Scout picks · head-to-head radar comparison · clickable analytics · public profile · direct messaging · talent search.

Owner — platform overview with 12 live stats · 14-day activity chart · complete athlete & coach tables · unified activity log. Hidden from every public list; access is decided entirely server-side; credentials live only in environment variables.

Privacy by design

The training video never leaves the device. Pose analysis runs entirely in the browser — only the resulting numbers are transmitted. This matters because most users are minors, and it also means the platform needs no server GPU: 10,000 athletes cost almost the same to serve as 10.

Every user can toggle research consent, export all their data as JSON, and delete their account permanently. Under-18 users require guardian consent (DPDP Act). Full Privacy Policy and Terms are published at `/privacy` and `/terms`.

Tech stack

Layer	Choice
Backend	Flask (Python) — 31 REST endpoints
Database	SQLite — 12 tables, cascading deletes
Frontend	Vanilla JavaScript (single file, no build step)

Computer vision	MediaPipe Pose Landmarker — on-device, 33 landmarks
Charts & PDF	Chart.js, jsPDF
Hosting	Render with gunicorn, auto-deploy on push
Extras	Progressive Web App · offline shell · 3-language UI (English / हिन्दी / Hinglish)

Languages: JavaScript · CSS · Python · HTML · SQL **Scale:** ~1,300 backend lines · ~3,800 frontend lines · 31 endpoints · 12 tables · 159 JS functions · 35 sports · 33 keypoints.

Security

Threat	Defence
SQL injection	All queries parameterised
Cross-site scripting	User text escaped before the DOM
CSRF	Token on every state-changing request, matched to session
Password theft	bcrypt, 12 rounds, unique salt per user
Session hijacking	HttpOnly + SameSite cookies; Secure in production
Broken access control	Role decorators on every route; identity from session, never URL

Admin credentials are stored in environment variables, never in source — which is why this repository can safely be public.

Running locally

```
# 1. clone
git clone https://github.com/manthandhanraj/AthletixAI
cd AthletixAI

# 2. install
pip install -r requirements.txt

# 3. (optional) create a .env for the owner account and email
#    OWNER_EMAIL=owner@example.com
#    OWNER_PASSWORD=your-strong-password

# 4. run
python app.py          # → http://127.0.0.1:5000
```

Demo logins: athlete `arjun@athletix.ai / 1234` · coach `coach@athletix.ai / 1234`

Honest limitations

Stated plainly, because a measurement tool that hides its limits can't be trusted.

1. **Rule-based, not trained.** The pose estimation is real ML; the scoring formulas encode domain knowledge — they are not learned weights.
2. **Not yet calibrated.** Proper calibration needs 100+ athletes' clips paired with measured performance (timed sprints, measured jumps, coach ratings).
3. **Relative, not absolute.** "Your speed improved 8% this month" is dependable; "your speed is exactly 82" is a calibrated estimate.
4. **2D pose has limits.** Depth is unreliable and camera angle matters, so the engine leans on symmetry and relative change over absolute distance.
5. **SQLite is single-writer.** Fine for hundreds of concurrent users; PostgreSQL is the planned next step.
6. **Movement, not tactics.** The engine reads movement quality; sport-specific technique judgement still needs a coach.

Roadmap

- **Now** — Cricket bowling module: arm angle vs the ICC 15° legality rule, run-up speed, front-foot landing, follow-through.
- **Next** — Calibration with real academy data; fit scoring weights to measured performance.
- **Then** — Per-sport models, one sport at a time — never claiming accuracy before the data supports it.
- **Later** — Academy dashboards, native mobile app, more regional languages.

The principle: **depth before width** — one sport done properly beats 35 done vaguely.

Credits

Built by **Archita Tripathi** and **Manthan Dhanraj**. Built with AI assistance — the architecture, product decisions, debugging and deployment are ours, and we can walk through any line of it.

Free for athletes. Always.